

Exercise 3: September 06, 2025

BIS 603/604: ERD

Your friends have asked you to design a simple database to help them with their new book rental business. The following are what they told you:

1. Your friend separates the type of books into 3 groups include fiction, education, and cartoon books. Each group has many genres such as
 - Fiction books: mystery, horror, history, science, etc.
 - Cartoon books: comedy, fantasy, action, horror, etc.
 - Education books: science, history, mathematics, etc.

Each time a new book is released, he will purchase several types of books and several copies of each type. All these books will buy for rental to customers. The minimum number of copies required for each genre of cartoon books is at least three or that up to five copies. Other types of books require at least two or that up to three copies for each genre.

2. Luckily with your friend, his mother gives him a new building for expand his rental book business. Therefore, he wants to keep fiction and cartoon books in the old building. The education books will be kept in the new building. Each type of books must be kept in each shelf. There are many shelves stored at the ground floor, second floor, and third floor of each building. If you have any idea in arranging all of the books for easy finding, please suggest to your friend.
3. In order to rent a book, customers have to become members first. The several type of member, platinum, gold, silver, can be classified based on the amount of cash deposit. They must give the cash deposit in the form of a credit card number or cash. Moreover, the usual address information (address, city, province, country, postal code), and a contact telephone number (area code, the phone number) are also provided. The member rate is different between each type of member. The silver member cannot rent the educational books.
4. When a copy of a book is rented out, the customer's details are recorded. At the same time, the date the book went out and the date it is expected returned to the store is recorded. The amount charged for the rental is also recorded.
5. On New Year day, there will be a promotion to the member. The type of promotion including (1) Rent 1 extra 1 (2) Extend the number of rental days when rented every 4 books. (3) Wave for the membership fee if rented on New Year day for 20 books.

What they want to achieve, among other things are to:

6. Identify the availability of a particular book title i.e. if all copies are out on rental. If not, how many copies are available for each type.
7. Determine the revenue generated from a particular book title.
8. Determine the revenue generated from each type of customer.
9. Identify the number of contacts from member and which promotion the member is interested in.